

FileEditSelectionViewGoRun

AI Lab Programs

AI_Lab_Programs

AI_Lab_Programs

.dist

.gitattributes

AI_Program_1

AI_Program_1(24BECS124).pdf

AI_Program_2

AI_Program_2(24BECS124).pdf

AI_Program_3

AI_Program_3(24BECS124).pdf

AI_Program_4

AI_Program_4(24BECS124).pdf

AI_Program_5

AI_Program_6

AI_Program_7

README.md

AI_Program_6

AI_Program_7

AI_Program_4

AI_Program_7 > ...

```
1 import math
2
3 def entropy(data):
4     labels = [row[-1] for row in data]
5     total = len(labels)
6
7     counts = {}
8
9     for label in labels:
10         counts[label] = counts.get(label, 0) + 1
11
12     return -sum(
13         (count / total) * math.log2(count / total)
14         for count in counts.values()
15     )
16
17 def split_data(data, attribute_index, value):
18     return [
19         row[:attribute_index] + row[attribute_index + 1:]
20         for row in data
21         if row[attribute_index] == value
22     ]
```

ProblemsOutputDebug ConsoleTerminalPorts

Code

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

Generated ID3 Decision Tree:

{'Outlook': {'Rain': 'Yes', 'Overcast': 'Yes', 'Sunny': 'No'}}

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

main* 110t 0 0 0

Ln 16, Col 1 Spaces: 4 UTF-8 CRLF {} Python Go Live



The image shows a code editor interface with a sidebar on the left containing a file explorer for a project named 'AI Lab Programs'. The main area displays a Python script 'AI_Program_7' with the implementation of the ID3 decision tree algorithm. The code defines a function 'id3' that takes 'data' and 'features' as input. It calculates the information gain for each feature and selects the one with the highest gain. The data is then split based on the selected feature, and the process is repeated recursively for each subset. The bottom panel shows a terminal window where the script is executed using 'python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_7"', resulting in the output: 'Generated ID3 Decision Tree: {\'Outlook\': {\'Rain\': \'Yes\', \'Overcast\': \'Yes\', \'Sunny\': \'No\'}}'. The status bar at the bottom indicates the current position in the file (Ln 62, Col 43) and other settings like 'Spaces: 4', 'UTF-8', 'CRLF', and 'Python'.

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AI_Program_4(24BECS124).pdf

AI_Program_5

AI_Program_6

AI_Program_7

README.md

AI_Lab_Programs

AI_Program_6 U

AI_Program_7 U

AI_Program_4

AI_Program_7 > id3

```
45 def id3(data, features):
70     # Get values of best feature
71     feature_values = set(
72         row[best_feat_idx]
73         for row in data
74     )
75     # Create branches
76     for val in feature_values:
77
78         subset = split_data(
79             data,
80             best_feat_idx,
81             val
82         )
83         tree[best_feature][val] = id3(
84             subset,
85             remaining_features
86         )
87     return tree
88 # Example Usage
89 if __name__ == "__main__":
90
```

Problems Output Debug Console Terminal Ports

Code + -

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>
python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_7"
Generated ID3 Decision Tree:
{'Outlook': {'Rain': 'Yes', 'Overcast': 'Yes', 'Sunny': 'No'}}
PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

Ln 87, Col 16 Spaces: 4 UTF-8 CRLF {} Python Go Live

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AI_Program_4(24BECS124).pdf

AI_Program_5

AI_Program_6

AI_Program_7

README.md

AI_Program_6 UAI_Program_7 UAI_Program_4

AI_Program_7 > id3

```
89 if __name__ == "__main__":
90
91     dataset = [
92         ['Sunny', 'Hot', 'High', 'Weak', 'No'],
93         ['Sunny', 'Hot', 'High', 'Strong', 'No'],
94         ['Overcast', 'Hot', 'High', 'Weak', 'Yes'],
95         ['Rain', 'Mild', 'High', 'Weak', 'Yes'],
96         ['Rain', 'Cool', 'Normal', 'Weak', 'Yes']
97     ]
98
99     features = [
100         'Outlook',
101         'Temperature',
102         'Humidity',
103         'Wind'
104     ]
105
106     tree = id3(dataset, features)
107
108     print("Generated ID3 Decision Tree:")
109     print(tree)
```

ProblemsOutputDebug ConsoleTerminalPorts

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs> python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_7"

Generated ID3 Decision Tree:
{'Outlook': {'Rain': 'Yes', 'Overcast': 'Yes', 'Sunny': 'No'}}
PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

Ln 87, Col 16Spaces: 4UTF-8CRLF[] PythonGo Live